

Technical Datasheet — VBF-T4R1-DSH-01

Field	Value	Source
Rated capacity (Ah)	5.0 nominal / 5.037 measured at 1C, 25 °C	parameter set `Nominal cell capacity [A.h]`; cell/r6_t2_1c_dfn.json
Nominal voltage / voltage window (V)	2.5 – 4.2 (midpoint 3.742 at 1C)	parameter set cut-offs; cell/r6_t2_energy.json:midpoint_voltage_v
Rated energy (Wh)	17.95 (simulated 1C discharge, 25 °C)	cell/r6_t2_energy.json:energy_wh (time integration of V·I)
Energy density (Wh/kg)	471.6	cell/r6_t2_energy.json:energy_density_wh_kg (contract mass formula, electrolyte excluded)
Volumetric energy density (Wh/L)	925.8	cell/r6_t2_energy.json:energy_density_wh_l
Maximum continuous discharge rate	1C (5 A) verified by simulation	run-pyamm 1C_discharge protocol
–20 °C capability	1C capacity retention 99.27 % (cold-soaked start, self-heating removed by h = 80)	cell/r6_t2_retention.json (capacity ratio 253.15 K / 298.15 K)
Fast-charge capability	4C/45 °C: +9.6 K temperature rise (T_max 327.7 K ≤ 333.15 K), no lithium plating (anode potential min +20.5 mV); charge is voltage-limited at 4.2 V after 377 s (runner-defined charge-step capacity 0.419 Ah at 4C rate)	cell/r6_t2_4c45C_dfn.json
Operating temperature range	Simulated: –20 °C to +45 °C ambient (253.15–318.15 K), plus 25 °C nominal; storage range not simulated	lowT_discharge / 4C_charge_45C / 1C_discharge protocols
Cycle life	Not simulated (requires aging model) — must not be inferred from this datasheet	protocol boundary
Safety determination	4C/45 °C: no plating, T_max within 333.15 K red line — pass (simulation)	cell/r6_t2_4c45C_dfn.json
DC resistance	4.06 mΩ (midpoint slope caliber per calc-energy)	cell/r6_t2_energy.json:dcr_ohm
Power density	26.9 kW/kg	cell/r6_t2_energy.json:power_density_w_kg
Dimensions	Electrode stack 65 mm × 1580 mm × 188.8 μm (unwound); cell envelope not provided (not modeled)	parameter set + cell/r6_t2_energy.json
Mass	38.07 g (electrolyte excluded) / 44.39 g incl. 6.33 g fill (literature density 1.2 g/cm³)	cell/r6_t2_energy.json:mass_kg + pore-volume calc

Notes: this datasheet is a virtual (simulation) release; physical-prototype values require cell fabrication and testing. The electrolyte transport values (κ 1.1 S/m, D_e 3e-10 m²/s, $t_{\blacksquare}$ 0.6) are design targets (LHCE-class idealization) — confirm by measurement before production.